Supplementary Proof Details

for *The complete equidistribution classification of length-three mesh patterns on layered permutations*

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This note records the two finite ingredients used at the end of the classification: the incidence data for the coordinate identifications and the exact separation of the resulting components. All identities valid for arbitrary host size, including the signature formula, the pointwise criterion, the kernel-degree theorem, the coordinate lemma, and the eighteen distributional bridges, are proved in the main paper.

S1 The last two finite quotients

After pointwise equality, reverse-complement, and the complete kernel-degree reduction, the main paper obtains 200 classes. Of these, 28 are the unique-support classes. Among the remaining 172 classes, 84 are isolated under coordinate relabelling, while the other 88 are joined by the 43 edges of Table S2. Seven edges close triangles already connected by two earlier edges, so the rank drop is 36 and

$$28 + 84 + (88 - 36) = 164.$$

The five all-size bridge families proved in the main paper give the inventory in Table S1.

Table S1: The eighteen effective bridges from 164 to 146 components.

mechanism	parameter range	effective joins
uniform cut shift	$(a, b) \in \{0, 1, 2\}^2$	9
exceptional rational identity	Class 79	1
non-overlapping contraction	$z \in \{0, 1, 2, 3\}$	4
separator complementation	$a \in \{0, 1, 2\}$	3
MacMahon bridge	major index–inversion	1
total		18

For every listed bridge, its two endpoints lie in distinct coordinate components, and no bridge becomes redundant after the preceding bridges are added. Thus all eighteen joins are effective and the resulting graph has $164 - 18 = 146$ components. This is the upper-bound partition used in the complete-classification theorem.

S2 The 43 coordinate edges

We use the notation $(r; z; E)$ from the main paper, omitting commas inside the block word, internal-channel word, and external-set word. Thus $(111; 012; 03)$ denotes $r = (1, 1, 1)$, $z = (0, 1, 2)$, and $E = \{0, 3\}$.

Every row of Table S2 is oriented. The appendix of the main paper proves, symbolically for arbitrary layer number m and arbitrary positive layer sizes, the identity

$$P_\Sigma(L) = P_{\Sigma'}(U_i(L))$$

displayed by that row. Thus the table records finite incidence data for all-size mathematical identities; it is not a list of bounded numerical tests.

Table S2: Explicit oriented coordinate edges. An entry $\Sigma \longrightarrow \Sigma'$ in the U_i column means $P_\Sigma(L) = P_{\Sigma'}(U_i(L))$ for every positive host composition L.

class	local edge	map	oriented signature identity
3	1–2	U_1	$(111; 000; 01) \longrightarrow (111; 000; 03)$
11	1–2	U_1	$(111; 001; 01) \longrightarrow (111; 010; 03)$
20	1–2	U_1	$(111; 002; 01) \longrightarrow (111; 020; 03)$
12	1–2	U_2	$(111; 001; 02) \longrightarrow (111; 001; 23)$
12	1–3	U_3	$(111; 001; 02) \longrightarrow (111; 010; 01)$
12	2–3	U_4	$(111; 001; 23) \longrightarrow (111; 010; 01)$
22	1–2	U_2	$(111; 002; 03) \longrightarrow (111; 002; 23)$
22	1–3	U_3	$(111; 002; 03) \longrightarrow (111; 020; 01)$
22	2–3	U_4	$(111; 002; 23) \longrightarrow (111; 020; 01)$

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Table S2 continued

class	local edge	map	oriented signature identity
43	1–2	U_2	$(111;012;03) \longrightarrow (111;102;23)$
43	1–3	U_3	$(111;012;03) \longrightarrow (111;021;01)$
43	2–3	U_4	$(111;102;23) \longrightarrow (111;021;01)$
13	1–2	U_8	$(111;001;0) \longrightarrow (111;010;0)$
21	1–2	U_8	$(111;002;02) \longrightarrow (111;020;02)$
23	1–2	U_8	$(111;002;0) \longrightarrow (111;020;0)$
44	1–2	U_8	$(111;012;0) \longrightarrow (111;021;0)$
35	1–2	U_5	$(111;011;01) \longrightarrow (111;101;01)$
37	1–2	U_6	$(111;011;12) \longrightarrow (111;101;02)$
38	1–2	U_6	$(111;011;1) \longrightarrow (111;101;0)$
65	1–2	U_6	$(111;022;23) \longrightarrow (111;202;03)$
90	1–2	U_6	$(111;122;23) \longrightarrow (111;212;03)$
102	1–2	U_6	$(111;222;23) \longrightarrow (111;222;03)$
42	1–2	U_5	$(111;012;01) \longrightarrow (111;102;01)$
42	1–3	U_1	$(111;012;01) \longrightarrow (111;021;02)$
42	2–3	U_7	$(111;102;01) \longrightarrow (111;021;02)$
46	1–2	U_6	$(111;012;13) \longrightarrow (111;102;03)$
46	1–3	U_{11}	$(111;012;13) \longrightarrow (111;021;23)$
46	2–3	U_7	$(111;102;03) \longrightarrow (111;021;23)$
48	1–2	U_9	$(111;012;3) \longrightarrow (111;102;3)$
63	1–2	U_9	$(111;022;13) \longrightarrow (111;202;13)$
67	1–2	U_9	$(111;022;3) \longrightarrow (111;202;3)$
92	1–2	U_9	$(111;122;3) \longrightarrow (111;212;3)$
53	1–2	U_7	$(111;102;02) \longrightarrow (111;021;12)$
55	1–2	U_7	$(111;102;0) \longrightarrow (111;021;2)$
59	1–2	U_3	$(111;022;03) \longrightarrow (111;022;01)$
59	1–3	U_2	$(111;022;03) \longrightarrow (111;202;23)$
59	2–3	U_{10}	$(111;022;01) \longrightarrow (111;202;23)$
87	1–2	U_3	$(111;122;03) \longrightarrow (111;122;01)$
87	1–3	U_2	$(111;122;03) \longrightarrow (111;212;13)$
87	2–3	U_{10}	$(111;122;01) \longrightarrow (111;212;13)$
79	1–2	U_1	$(111;112;01) \longrightarrow (111;121;02)$
81	1–2	U_1	$(111;112;0) \longrightarrow (111;121;2)$
80	1–2	U_{11}	$(111;112;03) \longrightarrow (111;121;23)$

S3 Finite separation of the 146 components

Proposition S1 (Pairwise separation)**.** *Choose one signature from each of the* 146 *components obtained in Sections S1–S2. Representatives can be chosen so that every two distinct representatives have different distribution polynomials at some host size* $n \le 7$. *Among the* $\binom{146}{2} = 10{,}585$ *unordered pairs, the numbers whose first separating sizes are* $4, 5, 6, 7$ *are, respectively,*

$$10{,}260, \qquad 302, \qquad 21, \qquad 2.$$

Consequently the 146 *components are pairwise distribution-inequivalent.*

Proof. For a signature Σ, substitute the occurrence formula from the main paper into

$$D_{\Sigma,n}(q) = \sum_{m=1}^{n} \sum_{\substack{\ell_1+\cdots+\ell_m=n \\ \ell_i \ge 1}} q^{P_\Sigma(\ell_1,\ldots,\ell_m)}.$$

For $n \le 7$ this is a finite sum of monomials with integer coefficients. Exact coefficient comparison gives the four numbers in the statement. The two pairs that require $n = 7$ are displayed below; all other pairs separate by $n \le 6$.

Put

$$\Sigma_A = (111;000;\{0,2\}), \quad \Sigma_B = (111;000;\{1,2\}).$$

Then

$$D_{\Sigma_A,7}(q) = 7 + 13q^5 + 5q^6 + 4q^7 + 18q^8 + 8q^9 + 2q^{10} + 7q^{12},$$
$$D_{\Sigma_B,7}(q) = 7 + 13q^5 + 4q^6 + 6q^7 + 17q^8 + 8q^9 + 2q^{10} + 7q^{12}.$$

For

$$\Sigma_C = (111;222;\{0,2\}), \quad \Sigma_D = (111;222;\{1,2\}),$$

one has

$$D_{\Sigma_C,7}(q) = 49 + 7q + 5q^2 + 2q^3 + q^5,$$
$$D_{\Sigma_D,7}(q) = 48 + 9q + 4q^2 + 2q^3 + q^5.$$

A difference at a single size rules out equidistribution. Therefore no two of the 146 components can merge. $\square$